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| **Assignment:** | Artificial Intelligence (AI) Internship - Primetrade.ai |

# Bitcoin Market Sentiment and Trader Performance Analysis

## Objective

The objective of this project was to analyze the relationship between Bitcoin market sentiment and trader performance using historical trading data and the Fear and Greed Index.

The goal was to uncover patterns in profitability, trading activity, and trader behavior under different market sentiment conditions and derive insights that could support smarter trading strategies.

## Datasets Used

**1. Historical Trader Dataset:** This dataset contained detailed trade level information including execution price, trade size, trade direction, transaction fees, timestamps, and closed profit and loss values.

**2. Fear and Greed Index Dataset:** This dataset contained daily Bitcoin market sentiment classifications including Extreme Fear, Fear, Neutral, Greed, and Extreme Greed.

## Data Preprocessing

Several preprocessing steps were performed before analysis. Timestamp columns were converted into datetime format, common date fields were created, and both datasets were merged using the date column. Missing values and datatype inconsistencies were checked to ensure data quality.

## Exploratory Data Analysis (EDA)

The analysis focused on understanding how market sentiment affects trader profitability, win rates, trading activity, trade sizes, transaction fees, and overall trading behavior. Multiple visualizations were created using Matplotlib and Seaborn for pattern discovery.

## Key Findings

1. Extreme Greed produced the highest average profitability among all sentiment categories.

2. Extreme Greed also showed the highest trader win rate, indicating stronger overall trading consistency.

3. Fear markets generated the highest trading activity, suggesting traders become more active during uncertain market conditions.

4. Extreme Fear showed relatively lower win rates despite moderate profitability, indicating that a few large profitable trades may compensate for many losing trades.

5. Larger trade sizes strongly increased transaction fees.

6. Closed PnL showed weak correlation with most numerical variables, suggesting profitability depends more on market conditions and strategy execution than simple trade size.

## Correlation Analysis Insights

The strongest observed correlation was between Size USD and transaction fees, with a correlation coefficient of approximately 0.75. Closed PnL showed weak linear relationships with most numerical variables, indicating that profitability is influenced by more complex market and behavioral factors.

## Conclusion

The analysis demonstrated that Bitcoin market sentiment significantly influences trader behavior and performance. Extreme Greed conditions resulted in the strongest profitability and highest win rates, while Fear conditions triggered the greatest trading activity.

These findings highlight the importance of incorporating sentiment indicators into trading strategy evaluation and risk management.

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